Trend Detective

NASA Space Apps Challenge 2026: Be An Earth System Trend Detective! | Kuala Lumpur | 14 November 2026

What was checked

The BM interface now translates the NASA POWER loading and short-record messages, the scan table, its statistical summary and multiple-testing explanation. Browser checks covered polar locations, open ocean, the date line, the late-starting haze record and a five-year record. Three Malaysian temperature records were compared with direct NASA POWER CSV downloads, and a 45-year MERRA-2 series was cross-checked against NASA POWER at Giovanni's reported grid point.

Data flow



English narration

A student chooses a variable, place and period. The browser requests NASA POWER monthly values, skips -999 missing values and averages the valid months into yearly values. Trend Detective calculates the linear rate, Sen's slope, Mann-Kendall test and slope t-test, then displays the chart, 95% confidence band and a plain-language explanation. All calculations run in the browser; there is no app server.

■■■■

学生选择变量、地点和时间范围后，浏览器直接向 NASA POWER 获取月度数据。程序跳过 -999 缺失值，把有效月份计算成年平均值，再计算线性趋势、Sen 斜率、Mann-Kendall 检验和斜率 t 检验。页面显示图表、95% 置信区间和通俗说明。计算在浏览器内完成，不需要应用服务器。

BM ringkas

Pengguna memilih pemboleh ubah, tempat dan tempoh. Pelayar mendapatkan data bulanan NASA POWER, mengetepikan nilai hilang -999, mengira purata tahunan dan ujian tren, kemudian memaparkan carta, julat keyakinan 95% dan penerangan bahasa mudah. Semua pengiraan berjalan dalam pelayar tanpa pelayan aplikasi.

Methods note: the app reports Mann-Kendall, Sen's slope and a slope t-test. It warns when year-to-year autocorrelation is strong; its p-values are not autocorrelation-corrected, so interpret those cases cautiously.

Browser edge-case checks

Manual checks on the live local index.html page and NASA POWER API. PASS means the page gave a valid result or a clear expected warning. WARN marks questionable source coverage. RESTORED records the requested rollback; the short-record stale-result behavior was not re-tested after the rollback.

Case	Inputs	Observed result	Status
North Pole	89°N; T2M; 1981-2025	Loads; +0.60 °C/decade, Mann-Kendall p < 0.001.	PASS
South Pole	89°S; T2M; 1981-2025	Loads; +0.48 °C/decade, p < 0.001.	PASS
Open ocean	0°N, 140°W; GWETTOP; 1981-2025	App reports no trend; source returns 1.0 every month at this ocean point.	WARN
Date line	0°N; 179°E and 179°W; T2M; 2° box	Both sides load; about +0.18 and +0.17 °C/decade.	PASS
Haze / aerosols	Klang point; AOD_55; requested 1981-2025	25 annual rows, 2001-2025; source metadata says starts in 2001.	PASS
Short record	Klang point; T2M; 2021-2025	BM warning is translated. Original stale-results behavior was restored at the user's request, so a previous result can remain visible.	RESTORED
BM scan	Klang point; scan 18 variables	Summary, table headers and false-alarm explanation are in BM.	PASS

Issue for Danny - open-ocean soil wetness

At 0°N, 140°W, NASA POWER supplies topsoil wetness (GWETTOP) = 1.0 for every month. Trend Detective treats it as valid and reports a flat/no-trend result. That is not meaningful soil-moisture information over ocean. Suggested action: apply a land/valid-data mask for soil variables or show “no land data here.” A ready-to-forward reproduction note is saved as docs/verification/danny-issue.md.

The issue note is prepared, but not sent: no Danny contact or team messaging channel was identified in this workspace.

NASA POWER data check

We downloaded the official monthly point CSV for each site and opened the same point/year case in Trend Detective. The app computes an annual mean from the twelve monthly values. Sample values below are shown to 0.01 °C.

Site / year	POWER 12-month mean	POWER ANN	App	App - month	App - ANN
Klang 1981	25.18	25.18	25.18	0.00	0.00
Klang 2000	25.17	25.17	25.17	0.00	0.00
Klang 2025	25.68	25.68	25.68	0.00	0.00
Penang 1981	25.76	25.76	25.76	0.00	0.00
Penang 2000	25.80	25.80	25.80	0.00	0.00
Penang 2025	26.49	26.50	26.49	0.00	-0.01
East Coast 1981	25.11	25.12	25.11	0.00	-0.01
East Coast 2000	25.15	25.15	25.15	0.00	0.00
East Coast 2025	25.69	25.69	25.69	0.00	0.00

Result: all 9 app values match the mean of NASA POWER's 12 downloaded monthly values at the app's 0.01 °C display precision. Two values differ from the separate POWER ANN field by only 0.01 °C. NASA documents annual values as an aggregation of monthly values; monthly CSV values are rounded for display, so the annual column can differ by one last-place digit.

Direct monthly CSV downloads

• Klang Valley (3.1 N, 101.6 E) - raw downloaded file saved in docs/verification/.

• Penang (5.35 N, 100.4 E) - raw downloaded file saved in docs/verification/.

• East Coast (5.4 N, 102.6 E) - raw downloaded file saved in docs/verification/.

POWER methods: Monthly API documentation · monthly-to-annual processing

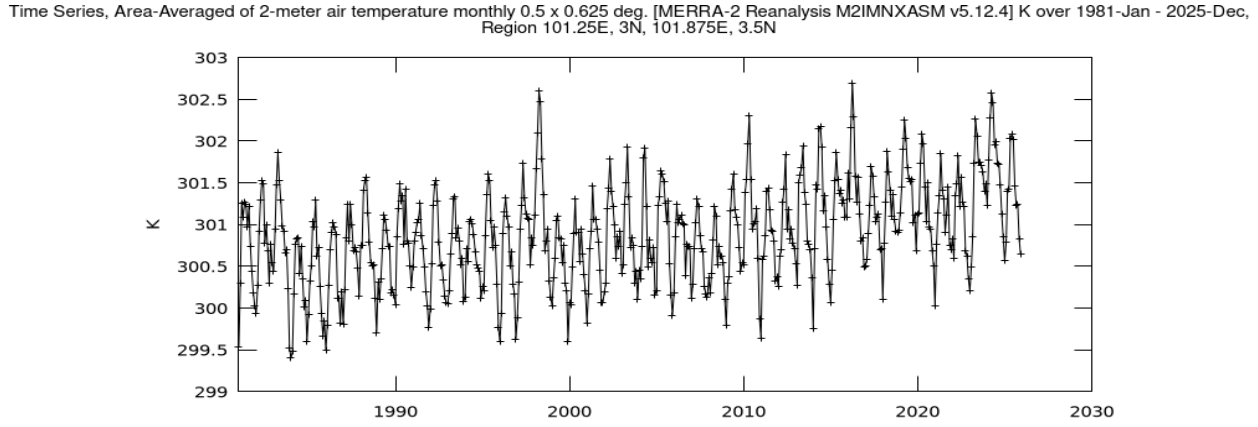
Independent Giovanni cross-check

Authenticated Giovanni v4.40 produced a monthly MERRA-2 2-meter air temperature series (M2IMNXASM_5_12_4_T2M) for January 1981 through December 2025. The selected bounds were 101.25°E-101.875°E, 3.0°N-3.5°N; Giovanni identifies the effective data grid as one point at 101.25°E, 3.0°N.

Annual summary	Giovanni (°C)	NASA POWER (°C)	Difference (°C)
1981 mean	27.535338	27.536667	-0.001328
2000 mean	27.461013	27.464167	-0.003154
2025 mean	28.157429	28.160000	-0.002571
OLS trend / decade	+0.186268	+0.186135	+0.000133

We compared all 540 overlapping monthly values with NASA POWER at the same grid point. Giovanni values were converted from K to °C; the monthly mean absolute difference is 0.0069 °C (maximum 0.0343 °C). After averaging each year's 12 months, the annual mean absolute difference is 0.0025 °C. The OLS slopes were calculated independently from those annual means; the Giovanni plot itself is a monthly series and does not display a trend test.

This is a same-grid cross-check of the NASA data path. The earlier Trend Detective Klang Valley spot check used 3.1°N, 101.6°E, which differs from Giovanni's effective point (3.0°N, 101.25°E); do not compare those separate locations as if they were the same point.



- The user-selected region was defined by 101.25E, 3N, 101.875E, 3.5N. The data grid also limits the analyzable region to the this point: 101.25E, 3N. This analyzable region indicates the spatial limits of the subsetted granules that went into making this visualization result.

Giovanni-generated plot: monthly 2-meter temperature in K, 1981-2025. The caption identifies the selected bounds and the single effective data-grid point.

Reproduce: Giovanni selection · same-grid NASA POWER CSV · MERRA-2 product · Giovanni User Guide.

Matched chart comparison: Klang Valley T2M

Same variable, point and years in both panels: monthly NASA POWER MERRA-2 T2M values, averaged by year, at 3.1°N, 101.6°E from 1981–2025. The chart above is the image saved from Trend Detective. The plot below is rebuilt from the official NASA POWER monthly CSV saved with this report; it is not a screenshot of the POWER website.

Trend Detective app chart export

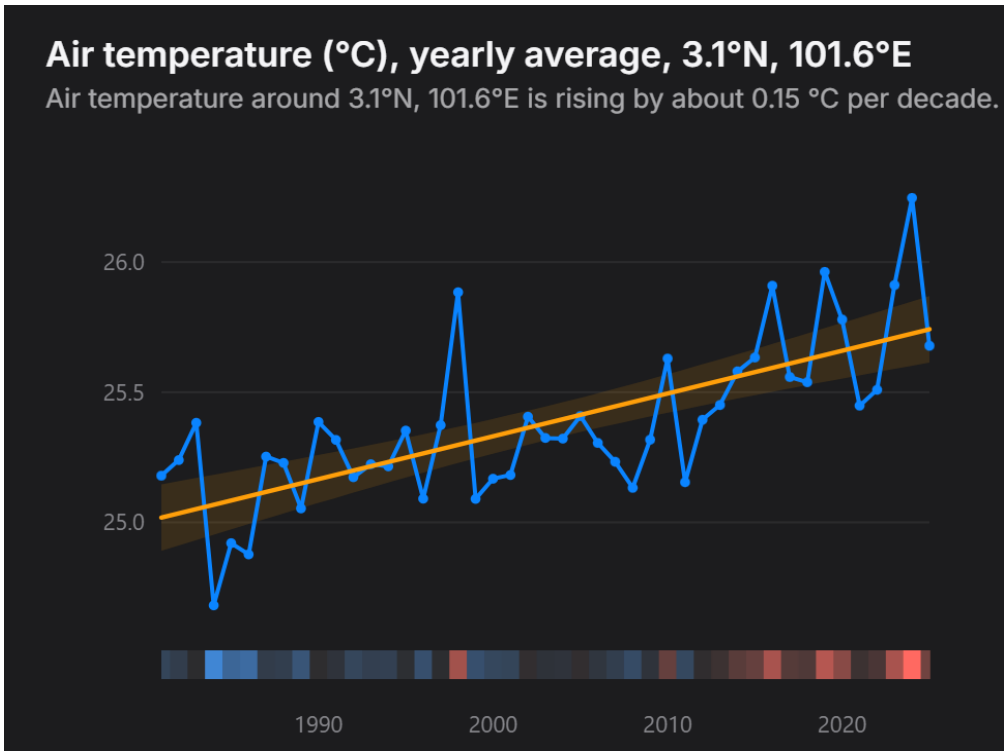
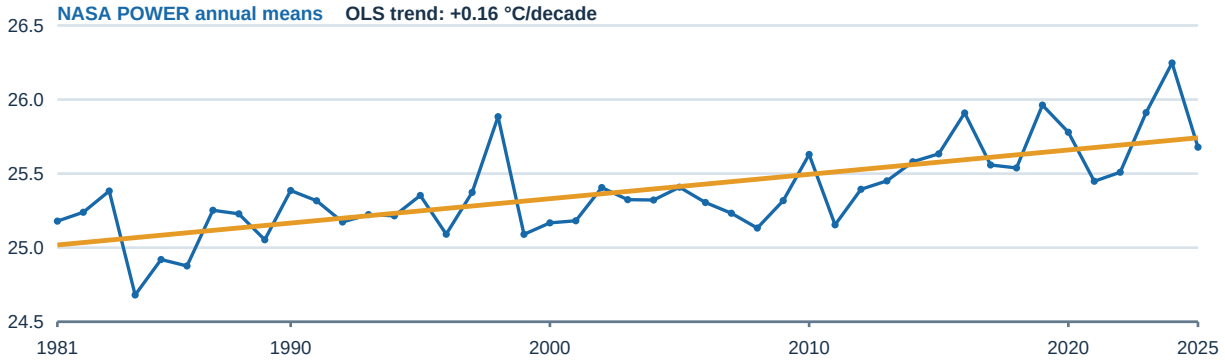


Chart panel cropped from the app's native PNG export; the original export remains in the project evidence. The app displays Sen's slope (+0.15 °C/decade). The independent NASA POWER plot below shows an ordinary least-squares slope (+0.16 °C/decade); the small difference is expected because these are different trend estimators.

NASA POWER monthly CSV, averaged and replotted



Source: NASA POWER monthly API request · downloaded response: docs/verification/power-t2m-klang-valley-1981-2025.csv. Monthly values marked -999 are omitted before annual averaging.